

Functional and Non-Functional Requirements List

1. Functional Requirements (FR)

- **FR01 - Real-Time Inventory Management:** The system must allow real-time stock control desglosed by size and product type.
- **FR02 - Production Module and Status:** The system must link production status with inventory availability to provide realistic delivery times.
- **FR03 - Product and Price Catalogue (CRUD):** The system must allow users to create, read, update, and delete (CRUD) products with their respective sizes and base prices (Life Jackets: Flea, Infant, Junior, Medium, Adult, Extra Large; Swimming Boards and Sliders).
- **FR04 - Quotation Automation:** The system must automatically calculate final prices, including VAT (16%), optional freight based on the delivery zone, and apply discounts if applicable.
- **FR05 - Document Generation (PDF):** The system must automatically export professional quotations in PDF format in less than 2 minutes.
- **FR06 - Customer Management:** The system must allow users to register customers and store their purchase history to facilitate recurring sales.
- **FR07 - Low Stock Alerts:** The system must generate automatic visual notifications when the inventory of any product runs low.
- **FR08 - Dashboard and Reports:** The system must provide a reporting interface to visualize the general status of the business.

2. Non-Functional Requirements (NFR)

- **NFR01 - Interface Usability (UX/UI):** The system must provide a simple and intuitive web-based interface optimized for warehouse and sales employees, incorporating KAVEM's corporate identity (company logo and color palette).
- **NFR02 - Response Time Efficiency:** The generation and processing of a complete quotation must not exceed 2 minutes.
- **NFR03 - Synchronization and Performance:** The processing and update time of inventory data must be reflected within a maximum of 15 minutes (ideally in real time to mitigate double-entry errors).
- **NFR04 - Architecture:** The system must be a web-based environment application.
- **NFR05 - Scalability (Quality Attribute):** The system design must lay the technical foundation for future electronic invoicing integration.

Constraints to Take into Account

Official Base Prices (VAT and Freight Not Included)

- **Life Jackets:** Flea (\$120), Infant (\$135), Junior (\$150), Medium (\$165), Adult (\$180), Extra Large (\$220).
- **Swimming Boards / Sliders:** Swimming Board One Size (\$100), Slider Small (\$120), Junior (\$190), Medium (\$250), Adult (\$300).

Taxes

- The mandatory applicable VAT is 16% on the base price.

Packaging Constraints (Capacities)

- A bag/package of swimming boards contains 10 to 20 units.
- A bag/package of life jackets contains 5 to 10 packed units.

Manufacturing Incomplete Goods & Materials (Raw Material to Control)

- Fabric rolls (1.5 mts wide x 50 mts long in different colors).
- Polypropylene straps in different colors.
- Samsonite-type buckles with safety locks.
- Thermoregulated polyethylene foam padding.
- Sewing supplies (Thread and needles).

Product States in Production

- Cut pieces, unscreenprinted boards, finished pieces (stitched life jacket / screenprinted board), and bagged pieces.